# Monte Carlo and sliced Wasserstein k-means clustering for regime detection

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# Research Motivations (1)

- Industry is in need of **interpretable Machine Learning** algorithms for detecting regimes in the market
- Detecting market regimes in **multi-dimensional time-series data**

# Research Motivations (2)

- Extend the paper "*Automated regime detection in multidimensional time series data using sliced Wasserstein k-means clustering*" by Luan et. Al (2025)
- Novel approach to regime detection using **Wasserstein k-mean clustering** to create clusters between probability measures.
- Does not use any **preconceived model of the distribution** of returns
- **Can be generalized to higher dimensional time-series** through sliced Wasserstein k-means clustering algo.

# Industry Background and Publications

- **Industry background**
  - Data Analytics (Munich Re)
- **Key Research works**
  - *Reinforcement Learning for continuous-time mean-variance portfolio optimization* (Technical University of Munich)
  - *Active Learning Techniques for Pomset Recognizers* (EPITA Research Laboratory & École Polytechnique)
    - Published research for the 29th International Conference on Foundations of Software Science and Computation
      Structures

# Research Objectives

- Use **computationally tractable** Monte Carlo algorithm to approximate the sliced Wasserstein k-means distance instead of using fixed predefined projection vectors.

- Be able to find **turning points** in the regime detection using implied probability approach. (idea using Gibbs distribution).

# Chapter 1 : Research Methodology and Outline

- Laying out the theoretical foundations ((sliced) Wasserstein distance, (Wasserstein barycenter, Stream, etc...)
  - -> can Monte Carlo create value in this aspect

- Proposing new algorithm with Monte Carlo (repu

- Testing
  - Testing new algorithm on **synthetic data**
  - Testing new algorithm on **real data (FX values, Commodities)**
  - Compare with original papers results + computational gains/losses using Monte Carlo
  - Develop an algo to model real-time regime-change in the markets.

# Chapter 2 : Literature Review

- "*Clustering Market Regimes using the Wasserstein Distance*" by Horvath et al. (2024)
  - introduced idea of using Wasserstein distance to do 1-dimensional time-series regime detection.

- "*Sliced and Radon Wasserstein barycenters of measures*" Rabing et al (2024)
  - Approximation of multidimensional Wasserstein distance using "sliced Wasserstein distance"

- "*Repulsive Monte Carlo on the sphere for the sliced Wasserstein distance*'" Petrovic et al (2025)
  - Introduced numerical scheme to use Monte Carlo and keep a numerically tractable algorithm Note : **Paper not mentioned by the original paper (Luan et al (2025)**

# Chapter 3: Theoretical Foundations

Wasserstein p-distance d=1:

$$\mathcal{W}_p^p(\mu,\nu) = \frac{1}{N}\sum_{i=1}^{N} |\mu_i^* - \nu_i^*|^p,$$

d>1 : Projection on d=1

More explicitly, given an empirical measure $\mu \in \mathcal{P}_p(\mathbb{R}^d)$, we can define a *projected* empirical measure $\mu'(\theta) \in \mathcal{P}_p(\mathbb{R})$, given by

$$\mu'(\theta) = \frac{1}{N}\sum_i \delta_{x_i'}, \tag{14}$$

where

$$x_i' = \langle x_i, \theta \rangle \tag{15}$$

Wasserstein Barycentre d= 1

$\mathcal{K} = \{\mu_m\}_{1\le m\le M}$, the Wasserstein barycentre $\bar{\mu}^{\mathcal{W}_p}$ of $\mathcal{K}$ is given by

$$\bar{\mu}^{\mathcal{W}_p} = (\bar{\mu}_1, \ldots, \bar{\mu}_N), \tag{11}$$

where, for $p = 1$,

$$\bar{\mu}_j^{\mathcal{W}_p} = \mathrm{Median}(\mu_{1,j}^*, \ldots, \mu_{M,j}^*) \quad \text{for } j = 1, \ldots, N \quad (p = 1), \tag{12}$$

and, for $p = 2$,

$$\bar{\mu}_j^{\mathcal{W}_p} = \mathrm{Mean}(\mu_{1,j}^*, \ldots, \mu_{M,j}^*) \quad \text{for } j = 1, \ldots, N \quad (p = 2) \tag{13}$$

Approximating the Wasserstein barycenter

$$\overline{\mathcal{W}}_p(\mu,\nu) \simeq \frac{1}{L}\sum_{l=1}^{L} \mathcal{W}_p(\mu'(\theta^l), \nu'(\theta^l)),$$

<u>IDEA</u> : Projection directions the L, thetas are randomly generated approximating the barycenter. (assuming ergodicity)

# Chapter 4: Testing MC sliced-Wasserstein k-means clustering (synthetic data)

Performance metric :
- **Total Accuracy (TA),**
- **Mean centroid-centroid distance,**
- **point-centroid distance**

$$\mathrm{TA}(\mathcal{C}, \tilde{t}) = \frac{\sum_{t_i \in \tilde{t}} \mathbb{I}_{\hat{y}_{t_i} = y_{t_i}}}{\sum_{t_i \in \tilde{t}} (\mathbb{I}_{\hat{y}_{t_i} = y_{t_i}} + \mathbb{I}_{\hat{y}_{t_i} \neq y_{t_i}})},$$

**Model:**
A) Monte Carlo Wasserstein k-clustering algorithm

**Experiment:** Simulate and compare 2D experiment with results in the Paper by Luan et al.

**Data: Synthetic Data**
Correlated Brownian Motions. (with multiple regimes bullish/bearish) (15- 20 years data)
- K =2 clusters (bullish/bearish)
- 2D and then 4D (ideally more)
- Correlation factors

# Chapter 5: Testing on real world data and results

**Performance metrics:**
- **Total Accuracy (TA)**
- **Mean centroid-distance**
- **Point centroid-distance**

**Model:**
A) sliced Wasserstein k-clustering

B) Monte Carlo Wasserstein k-clustering algorithm

**Experiment:**
Compare performance between Model A and B for 4D to 5D time series data.

**Data: (15-20 years data)**
**- FX Spot Time Series Data (4D):**
- AUDUSD, NZDUSD, USDCAD, USDNOK (Commodity Currency Cluster)
- EURUSD, GBPUSD, USDJPY, USDCHF (G4 majors)

**Commodities (4D- 5D) :**
- WTI Crude Oil, Brent Crude, Natural Gas, Heating Oil.

- Copper, Aluminum, Zinc, Nickel, Lead. (5D) (interesting to see on very recent data)

# Chapter 6: Future works and experiments on implied probability

If you have $K$ regimes with centroids $\bar{\mu}_1, \bar{\mu}_2, ..., \bar{\mu}_K$ and a new distribution $\nu$. The probability that distribution $\nu$ belongs to regime $k$ is given by the Gibbs distribution:

$$P(k|\nu) = \frac{\exp(-\beta \cdot \text{dist}(\nu, \bar{\mu}_k))}{\sum_{j=1}^{K} \exp(-\beta \cdot \text{dist}(\nu, \bar{\mu}_j))} \tag{1}$$

where:

- $\text{dist}(\nu, \bar{\mu}_k) = \overline{\mathcal{W}}_p(\nu, \bar{\mu}_k)$ is the Sliced Wasserstein distance between the observation distribution and the $k$-th regime centroid.
- $\beta$ is the sensitivity parameter (inverse temperature) that controls the "softness" of the regime assignment.
- $K$ is the total number of identified cluster centroids (regimes).

# Contributions to Science (academic)

- Contribute to academic literature on regime detection and clustering.

- Extension of suggestion offered by the paper "Automated regime detection in multidimensional time series data using sliced Wasserstein k-means clustering"

- Advancing our understanding of computationally tractable algorithms for regime detection

# Impact Statement (business/industry)

- Before sliced- Wasserstein k-means clustering practically used only on 2D real-world financial time series (publicly available FX spot rate) and up to 3D synthetic data.

- Use Monte Carlo to be able to use it on more than 3D time-series.